

Program 10

Demonstrate kubernetes Dashboards and CLI for current monitoring

Structure of Program:

deployentes.yaml
service.yaml

Steps:

1. Start the minikube by setting cpus and memory

```
1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-10$ minikube start --cpus=2 --memory=4096
🐳 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
⚠️ You cannot change the memory size for an existing minikube cluster. Please first delete the cluster
👍 Starting "minikube" primary control-plane node in "minikube" cluster
🚜 Pulling base image v0.0.48 ...
👤 Updating the running docker "minikube" container ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ■ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
   ■ Using image docker.io/kubernetesui/dashboard:v2.7.0
   ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
   ■ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌞 Enabled addons: storage-provisioner, metrics-server, default-storageclass, dashboard
🏁 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

2. Write the deployments.yaml file

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

3. Write the service.yaml file

```

apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort

```

4. Apply the files

```

_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-10$ kubectl apply -f deployments.yaml
deployment.apps/nginx-deployment created
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-10$ kubectl apply -f service.yaml
service/nginx-service created

```

5. Verify the pods with CLI

```

_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-10$ kubectl get pods

```

NAME	READY	STATUS	RESTARTS	AGE
mycron-29434960-jkvvk	0/1	Completed	0	5h35m
mycron-29435290-vvnjh	0/1	Completed	0	3m39s
mycron-29435295-xzjj9	0/1	Completed	0	40s
nginx-deployment-6f9664446b-dfglq	1/1	Running	0	17s
nginx-deployment-6f9664446b-hn7s7	1/1	Running	0	18s

6. Verify the nodes

```

_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-10$ kubectl get nodes

```

NAME	STATUS	ROLES	AGE	VERSION
minikube	Ready	control-plane	23d	v1.34.0

7. Verify the services

```

_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-10$ kubectl get svc
NAME                TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
kubernetes           ClusterIP      10.96.0.1        <none>           443/TCP          6d23h
nginx-service        NodePort       10.102.80.253    <none>           80:30095/TCP     32s

```

8. Verify the pods log

```

_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-10$ kubectl logs nginx-deployment-6f9664446b-dfglq
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/12/19 04:15:40 [notice] 1#1: using the "epoll" event method
2025/12/19 04:15:40 [notice] 1#1: nginx/1.29.4
2025/12/19 04:15:40 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/12/19 04:15:40 [notice] 1#1: OS: Linux 6.14.0-33-generic
2025/12/19 04:15:40 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/12/19 04:15:40 [notice] 1#1: start worker processes
2025/12/19 04:15:40 [notice] 1#1: start worker process 29
2025/12/19 04:15:40 [notice] 1#1: start worker process 30
2025/12/19 04:15:40 [notice] 1#1: start worker process 31
2025/12/19 04:15:40 [notice] 1#1: start worker process 32
2025/12/19 04:15:40 [notice] 1#1: start worker process 33
2025/12/19 04:15:40 [notice] 1#1: start worker process 34
2025/12/19 04:15:40 [notice] 1#1: start worker process 35
2025/12/19 04:15:40 [notice] 1#1: start worker process 36

```

9. Describe the pod

```

Events:
Type      Reason      Age   From          Message
----      -
Normal    Scheduled   87s   default-scheduler   Successfully assigned default/nginx-deployment-6f9664446b-dfglq to minikube
Normal    Pulling     87s   kubelet          Pulling image "nginx:latest"
Normal    Pulled      71s   kubelet          Successfully pulled image "nginx:latest" in 3.717s (15.551s including waiting)
Normal    Created     71s   kubelet          Created container: nginx
Normal    Started     71s   kubelet          Started container nginx

```

```

_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-10$ kubectl describe pod nginx-deployment-6f9664446b-dfglq
Name:          nginx-deployment-6f9664446b-dfglq
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Fri, 19 Dec 2025 09:45:24 +0530
Labels:        app=nginx
               pod-template-hash=6f9664446b
Annotations:   <none>
Status:        Running
IP:            10.244.0.162
IPs:
  IP:          10.244.0.162
Controlled By: ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID:  docker://f25a71df7de0f7bcf1a548b5b5462d691e93f877f88494cf6b38b76c4920ce81
    Image:         nginx:latest
    Image ID:      docker-pullable://nginx@sha256:fb01117203ff38c2f9af91db1a7409459182a37c87cced5cb442d1d8fcc66d19
    Port:         80/TCP
    Host Port:     0/TCP
    State:         Running
      Started:     Fri, 19 Dec 2025 09:45:40 +0530
    Ready:         True
    Restart Count: 0
    Environment:   <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-bg2jl (ro)
Conditions:
  Type                               Status
  PodReadyToStartContainers         True
  Initialized                       True
  Ready                             True
  ContainersReady                   True
  PodScheduled                      True
Volumes:
  kube-api-access-bg2jl:
    Type:              Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:      kube-root-ca.crt
    Optional:           false

```

10. Open the application via minikube and view the same on browser

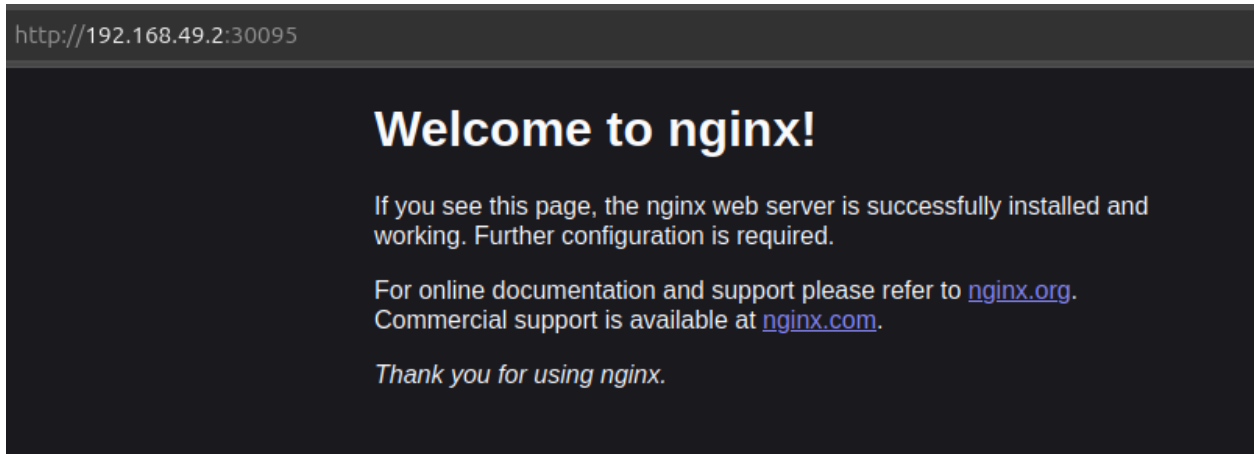
```

_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-10$ minikube service nginx-service

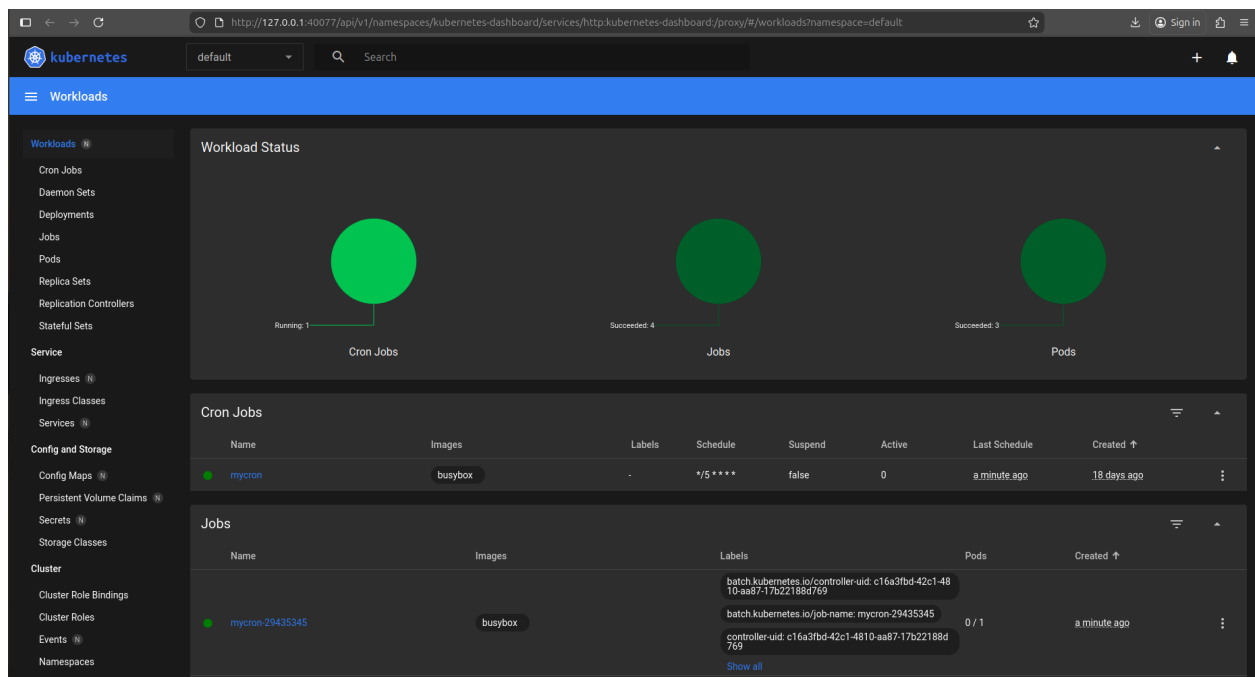
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:30095

 Opening service default/nginx-service in default browser...



11. Open the kubernetes dashboard and view the details



12. Delete the pods

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-9$ kubectl delete pods --all
pod "mycron-29435315-pnbrk" deleted from default namespace
pod "mycron-29435320-c4ggv" deleted from default namespace
pod "mycron-29435325-4br58" deleted from default namespace
pod "nginx-deployment-6f9664446b-dfglq" deleted from default namespace
pod "nginx-deployment-6f9664446b-hn7s7" deleted from default namespace
```

13. Delete the services

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-9$ kubectl delete svc --all
service "kubernetes" deleted from default namespace
service "nginx-service" deleted from default namespace
```

14. Delete the deployments

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-9$ kubectl delete deployments --all
deployment.apps "nginx-deployment" deleted from default namespace
```